

MonteCalc

User Handbook

Monte Carlo simulation for LibreOffice Calc and Microsoft Excel
following the MC FLO workflow

Version 0.9.5 · November 2026

LibreOffice extension (.oxt) · Excel add-in (.xll, 64-bit and 32-bit)

Contents

1 Introduction 4

1.1 Two editions, one program 4

1.2 What you can do with MonteCalc 4

2 Installation 5

2.1 LibreOffice extension 5

2.2 Excel add-in 5

2.3 Language 5

2.4 AI mentor (optional) 6

3 Quick start: a product launch 6

4 Concepts 7

4.1 Inputs, outputs and correlations 7

4.2 Arguments of the distributions 8

4.3 Modes of the spreadsheet 8

5 Building a model 8

5.1 Choosing a distribution: the wizard 8

5.2 Fitting distributions to data 9

5.3 Correlations 9

5.4 Outputs 10

5.5 Register, check and lock 10

5.6 Multiple simulation (variants) 11

6 Running a simulation 11

6.1 Settings 11

6.2 Start, preview, pause and stop 12

6.3 Memory Calculation and multiple cores 13

6.4 Seeds and reproducibility 14

7 Analysing the results 14

7.1 The result window 14

7.2 Result functions in your cells 16

7.3 Analysing a single iteration 17

7.4 The result workbook 17

7.5 Influence diagram 17

7.6 AI mentor 18

8 Interface reference 18

8.1 Excel: the MC FLO ribbon tab 18

8.2 LibreOffice: sidebar, menu, toolbar and tab 19

8.3 Keyboard . . . 19
9 Differences between the editions . . . 20
10 Troubleshooting . . . 20
Appendix A Function reference . . . 22
Model functions: continuous distributions . . . 22
Model functions: discrete distributions . . . 23
Model functions: time series . . . 23
Model functions: output, correlation, bounds . . . 24
Result functions (need a simulation) . . . 24
Information and settings . . . 26
Data tools . . . 26
Regression . . . 27
Bayesian statistics . . . 28
Batch forecasting . . . 29
Matrix functions . . . 29
Memory Calculation vector functions . . . 30
Appendix B Files and names . . . 31

1 Introduction

MonteCalc adds Monte Carlo simulation to spreadsheets. You describe the uncertain inputs of a model with probability distributions, mark the results you are interested in, and MonteCalc recalculates the model thousands of times. The result is not a single number but a distribution: how likely a loss is, which input drives the risk, what the worst and best cases are.

MonteCalc follows the workflow, function names and conventions of **MC FLO** (user guide and wiki on www.mcflosim.ch). Models written for MC FLO calculate in MonteCalc. MonteCalc is an independent implementation; it contains no MC FLO code and is not affiliated with MC FLOsim. The name MC FLO is used only to describe compatibility.

1.1 Two editions, one program

MonteCalc comes in two editions built from the same source code. The functions, the windows, the commands and the results are the same; a model gives the same results for the same seed in both.

	LibreOffice Calc	Microsoft Excel
File	`MonteCalc-windows_x86_64.oxt` / `MonteCalc-linux_x86_64.oxt`	`MonteCalc-Excel-x64.xll` (64-bit Office) / `MonteCalc-Excel-x86.xll` (32-bit Office)
Requires	LibreOffice 7.6 or later (64-bit)	Excel 2010 or later, Windows
Commands	MC FLO sidebar deck, MC FLO menu and toolbar, Extension tab (tabbed interface)	MC FLO ribbon tab
Argument separator	`;` (e.g. `=FLOsimula_Normal(100;15;"Demand")`)	`,` in English Excel, `;` in German or Spanish Excel
Result workbook	`.ods`	`.xlsx`

In this handbook formulas are written with `;` as in LibreOffice and in German or Spanish Excel. In English Excel type `,` instead.

1.2 What you can do with MonteCalc

- **Model uncertain inputs** with 33 distributions (20 continuous, 8 discrete, 5 time series, including your own data), optionally truncated, rounded or summed.
- **Correlate inputs** with nine copula types (normal, Clayton, Gumbel, Student-t and asymmetric types), calibrated to the rank correlation you ask for.
- **Simulate** thousands of iterations, with a live preview window, pause and stop, and Memory Calculation on all processor cores.
- **Analyse** results in the result window (histogram, cumulative curve, tornado chart, sensitivities, scatter, decision matrix, statistics), with more than 100 `fMC_` result and tool functions in your own cells, in a result workbook and in the influence diagram.
- **Compare variants** of a decision with multiple simulation.
- **Fit** distributions and time-series models to historical data and forecast with Holt-Winters.
- **Ask the AI mentor** about your model and its results.
- Work in **English, German or Spanish**.

2 Installation

2.1 LibreOffice extension

1. Download `MonteCalc-windows_x86_64.oxt` (Windows) or `MonteCalc-linux_x86_64.oxt` (Linux) from the latest release.
2. In LibreOffice choose **Tools → Extension Manager… → Add…**, select the file and accept. Alternatively double-click the .oxt file.
3. **Restart LibreOffice completely.** On Windows also close the Quickstarter in the system tray. Until LibreOffice has restarted, the MonteCalc functions show `#NAME?`.
4. Open `business_case.ods` (attached to every release). You find the **MC FLO** sidebar deck (icon at the right edge, or View → Sidebar), the **MC FLO** menu and toolbar, and the **Extension** tab of the tabbed interface (View → User Interface → Tabbed).

To update, remove the old version in the Extension Manager, add the new one and restart LibreOffice. To uninstall, select MonteCalc in the Extension Manager and choose **Remove**.

2.2 Excel add-in

1. Find out whether your Office is 64-bit or 32-bit: File → Account → About Excel; the first line ends in “64-bit” or “32-bit”. Download `MonteCalc-Excel-x64.xll` for 64-bit or `MonteCalc-Excel-x86.xll` for 32-bit Office.
2. Windows may block downloaded add-ins: right-click the file, choose Properties, tick **Unblock** and click OK.
3. In Excel go to **File → Options → Add-ins**, choose Manage: Excel Add-ins → Go… → Browse…, select the .xll file and confirm. The add-in now loads every time Excel starts.
4. Excel shows the **MC FLO** ribbon tab. Open `business_case.xlsx` and click **Start** in the Simulation group.

No Visual C++ runtime or other installation is needed; the add-in registers its ribbon for the current user only (no administrator rights). To update, replace the .xll file and restart Excel. To remove it, untick it in the Add-ins dialog.

If the MC FLO tab does not appear: open File → Options → Add-ins → Manage: COM Add-ins → Go… and tick **MonteCalc**. All commands are also on the **Add-ins** tab (MC FLO menu) as a fallback.

2.3 Language

Windows, dialogs, messages, ribbon labels and chart captions are available in English, German and Spanish. **Auto** follows the language of LibreOffice or of Windows. To choose another language use **Settings… → Language** (LibreOffice) or **Other → Language** on the Excel ribbon. The choice is stored in `Documents\MC_FLO_language.txt` and shared by both editions.

Function names (`FLOsimula_…`, `fMC_…`), variable names and the sheets of the result workbook are the same in every language, so a model works in every language version.

2.4 AI mentor (optional)

The AI mentor needs an Anthropic API key. Put it into `Documents\MC_FLO_API.txt` or type it into the mentor window. The mentor sends the structure and the statistics of your model, never the cell contents.

3 Quick start: a product launch

The example workbook `business_case` (.ods for LibreOffice, .xlsx for Excel, attached to every release) evaluates the launch of a new product over five years. Open it and follow these steps.

The inputs (cells B4 to B10)

Cell	Input	Formula
B4	Market size (units / year)	`=FLOsimula_PERT(200000;350000;250000;"MarketSize")`
B5	Market share	`=FLOsimula_Triangular(0.05;0.12;0.08;"Share")`
B6	Price per unit	`=FLOsimula_Normal(25;2;"Price";;18;32)` (normal, truncated to 18…32)
B7	Unit cost	`=FLOsimula_Triangular(12;18;14;"UnitCost")`
B8	Investment	`=FLOsimula_LogNormal(0;12.8992;0.1492;"Investment")`
B9	Launch delayed (1 = yes)	`=FLOsimula_Bernoulli(0.2;"Delay")`
B10	Annual market growth	`=FLOsimula_Normal(0.03;0.02;"Growth")`

Between simulations each input cell shows its expected value, so the spreadsheet stays a normal, readable model.

Correlations, outputs and results

- `=FLOsimula_correlation("Share";"Price";-0.6;0)` in B13: a higher price means a lower market share.
- `=FLOsimula_correlation("UnitCost";"MarketSize";-0.4;0)` in B14: economies of scale.
- The cash flow table (rows 22–27) and the net present value (B29) are ordinary spreadsheet formulas.
- `=B29+FLOsimula_output("NPV")` in B30 and the total cash flow in B31 mark the outputs.
- Rows 34–42 use result functions, e.g. `=fMC_Average("NPV")`, `=fMC_Percentile("NPV";fMC_Worst())` and `=fMC_Bayes("NPV";">";0;"Price";">=";25)`.

Run it

1. Select cell B30 (the NPV output), so that the preview shows the NPV.
2. Click **Start** (Simulation group). The preview window opens and the histogram grows as the iterations come in.
3. When the run is finished, the result window opens for the NPV and rows 34–42 show the results.
4. Choose **Tornado chart** in the View list to see which inputs drive the NPV; click a bar for the scatter of that input.
5. Enter `0` as Magic stripe to read the probability of a negative NPV off the cumulative curve.

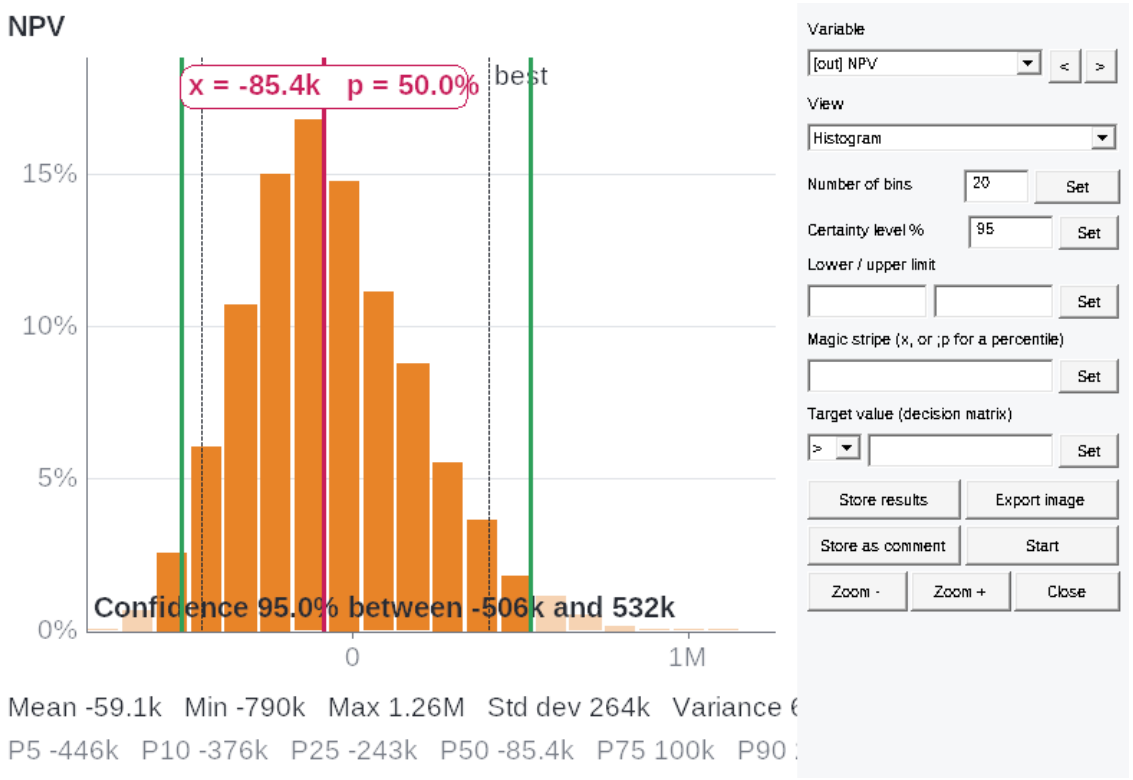


Figure 1 – Result window: histogram of the NPV with the magic stripe (red), worst and best case (dashed) and the 95% certainty range (green)

4 Concepts

4.1 Inputs, outputs and correlations

Kind	Written as	Named range	Cell colour
Input variable	`=FLOsimula_<Distribution>(parameters ; "Name"; …)`	`FLO_i_Name`	blue
Output variable	`=<formula>+FLOsimula_output("Name")`	`FLO_o_Name`	orange
Correlation	`=FLOsimula_correlation("A"; "B"; r; type)`	`FLO_c_A_B`	yellow
Locked input	an input fixed at its expected value (Lock)	–	red
Variant reference	rows Variant | Value | Reference (Multiple simulation)	–	grey

Inputs draw a new value in every iteration. **Outputs** are recorded in every iteration; `FLOsimula_output` adds 0, so `=B29+FLOsimula_output("NPV")` shows the value of B29. **Correlations** link two named inputs.

An MC FLO function must be the **principal function of its cell**: MonteCalc records the value of the cell, so `=FLOsimula_Normal(100;15;"Demand")*2` records the doubled value under the name Demand.

4.2 Arguments of the distributions

All input functions take their parameters first, then the name and four optional settings:

```
=FLOsimula_<Distribution>(parameters…; Variable_name; [Number]; [Min]; [Max]; [Round])
```

Argument	Meaning
Variable_name	Name of the input, used in correlations, result functions and charts. Without a name the input is drawn independently and Register variables can name it later.
Number	Number of occurrences per iteration; the draws are summed (compound distribution). Empty or 0 = 1.
Min, Max	Truncate the distribution to this range.
Round	Number of decimals (TRUE = whole numbers).

Example: `=FLOsimula_Normal(25;2;"Price";;18;32)` is a normal distribution with mean 25 and standard deviation 2, truncated to 18…32 (Number left empty).

4.3 Modes of the spreadsheet

Mode	The input cells show
Expected values (normal state)	the expected value of each distribution
During a simulation	the values of the current iteration
Preview mode (seed −1)	a new random value on every recalculation
Analyse iteration (Replay)	the input values of one stored iteration, written into the cells; **Expected values** restores the formulas

5 Building a model

5.1 Choosing a distribution: the wizard

Select the cell for the input and choose a distribution from **Choose continuous variable**, **Choose discrete variable** or **Choose time series**, or open the **Wizard**. The wizard shows the distribution with a sample, its statistics and the parameters:

- Change the parameters and click **Show** to redraw. Number of iterations / bins controls the sample.
- Drag the red **magic stripe** over the chart, or enter `x` (a value) or `;p` (a percentile) to read probabilities.
- **Load real data** overlays historical data from the selection; **Custom defined (from selection)** builds a `FLOsimula_UserDefined` distribution from the selected data.
- Enter the **Name of distribution** and click **Save into the selected cell**. MonteCalc writes the formula, creates the `FLO_i_` name and colours the cell.

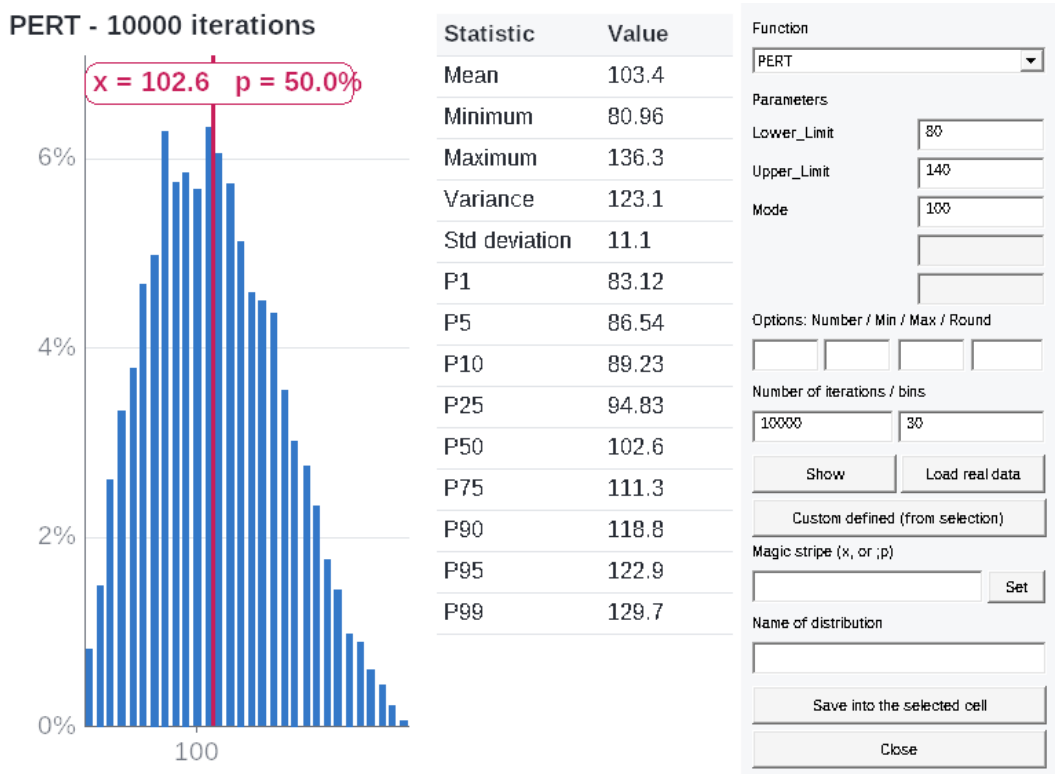


Figure 2 – The wizard with a PERT distribution (80, 140, mode 100)

Changing a variable. Select a cell that already holds an input and open the wizard: it shows the cell's distribution, parameters, name and options. **Update the selected cell** replaces only the `FLOsimula_` call, so the rest of the formula (e.g. `*2`) stays. Choosing another distribution from the menu keeps the name and the options.

Numbers may be typed with a decimal comma (`0,5`); the wizard writes `0.5` into the formula.

5.2 Fitting distributions to data

Select a column of historical data and choose **Fit distribution**. MonteCalc fits all suitable distributions by least squares against the empirical distribution function and ranks them with the Kolmogorov–Smirnov test and the AIC; a Q-Q plot shows the quality of the selected fit. Insert formula into current cell writes the chosen distribution. For data over time, Fit time series compares AR, MA, ARMA, ARCH and Wiener models (with Theil's U and a Holt-Winters comparison).

5.3 Correlations

Select an empty cell and choose **Insert correlation**. Pick the two inputs, type the coefficient or use the slider, and choose the correlation type. The scatter shows 1,000 sample points: both values beyond the 90% band green (upper) or red (lower), one value beyond it black, the rest blue. The rank correlation actually reached is shown in grey under the coefficient. **OK** writes the formula, for example `=FLOsimula_correlation("Price";"Investment";0.6;3)`.

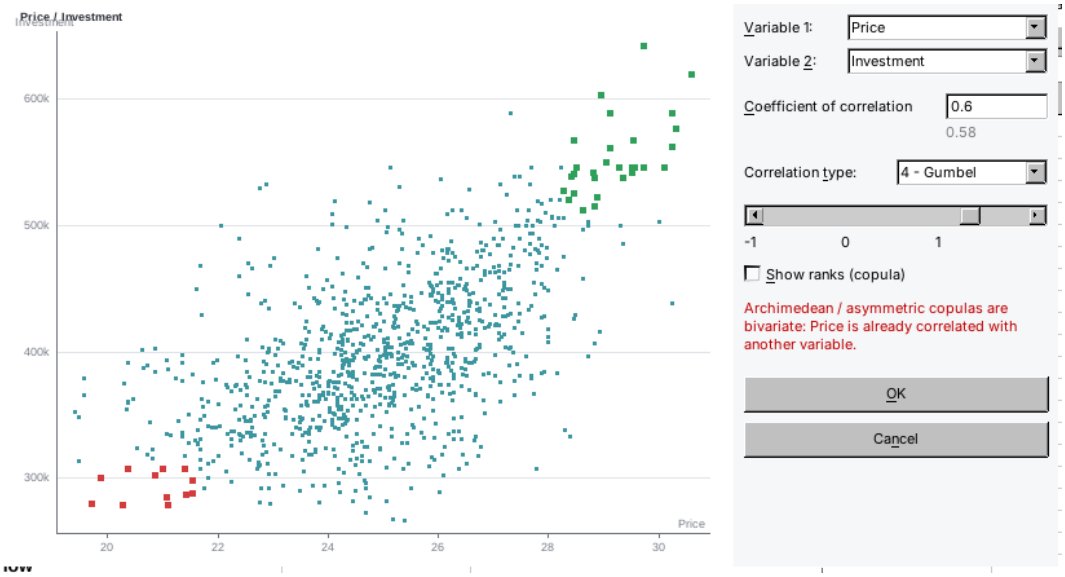


Figure 3 – Correlation window: Price and Investment, Gumbel copula, r = 0.6

Code	Type	Character
0	Normal	symmetric; the default
1	Clayton	strong joint movement in the lower tail
2	rClayton (rotated)	strong joint movement in the upper tail
3	Gumbel	upper tail dependence
4	rGumbel (rotated)	lower tail dependence
5	Student-t	joint extremes in both tails
6	asymmetric Gumbel	asymmetric (Khoudraji) Gumbel
7	asymmetric Triangular	asymmetric (Khoudraji) Clayton
8	asymmetric Fréchet	Marshall–Olkin type

The coefficient is calibrated so that the simulated **rank correlation** matches it. Inconsistent correlation matrices are repaired with a warning. Archimedean and asymmetric types (1–4, 6–8) are bivariate: an input can be part of only one such correlation, and a pair can be correlated only once; the window checks both.

5.4 Outputs

Select the cell with a result and choose **Insert output variable**. MonteCalc asks for a name (the label to the left is proposed) and turns `=B29` into `=B29+FLOsimula_output("NPV")`.

5.5 Register, check and lock

Command	What it does
Register variables	Names unnamed inputs (from the label to the left), creates the `FLO_i_` / `FLO_c_` / `FLO_o_` names and colours the cells.
Model	Brings a changed model up to date: variables that were overwritten or deleted are removed and every reference to them is replaced by 0; then the variables are listed.
Show variables	Lists all inputs, outputs and correlations with their cells and statistics; Details opens the result window.
Lock / Unlock	Fixes the selected input at its expected value for the next runs (red cell), e.g. to see its effect.

Command	What it does
Excel-Check (model check)	Writes a check workbook: volatile and lookup functions, MC FLO functions that are not the principal function of their cell, constants in formulas, empty references, unnamed inputs, duplicate names, unknown correlation partners.
FLO_Context sheet	A sheet describing every variable with the cells it depends on.

5.6 Multiple simulation (variants)

To compare decisions, enter rows of Variant | Value | Reference (the reference is the cell that takes the value, e.g. a price), select them and choose **Multiple simulation**. Each Start then simulates every variant. In the result window a variant list appears at the top and the **Variants** view overlays their distributions; result functions take the variant as optional last argument: `=fMC_Average("NPV";"High")`. Choose **Multiple simulation** on an empty cell to reset.

6 Running a simulation

6.1 Settings

The Simulation group holds the most used settings (Best-case, Real-case, Worst-case, Iterations and check boxes); **Settings…** holds all of them. They are stored in the workbook.

Setting	Meaning	Default
# iterations	Number of iterations of a run.	1000
Seed	0 = a new random sequence each run; any other number repeats the same iterations; −1 = preview mode (every recalculation draws random values).	0
Worst / Real / Best case level	Percentiles used for the worst, real and best case (result functions, charts, summary).	0.05 / 0.5 / 0.95
Histogram bins	Default number of bins of the histograms.	20
Use correlations	Off: all inputs are simulated independently (to see the effect of the correlations).	on
Memory Calculation	Evaluate the model with MonteCalc's own formula engine, on all processor cores (6.3).	off
Preview window	Show the live preview during the run (6.2).	on
Visualize simulation	Let the spreadsheet show every iteration (slow).	off
Store results in a result workbook	Write the result workbook after each run (7.4).	off
Use DIP	Store the outputs as `FLO_r_` arrays in the model.	off
Show the result window when a variable cell is selected	After a run, selecting a variable cell shows it in the result window, with the wedge to the cell (7.1).	on
Language	Auto, English, Deutsch, Español.	Auto

6.2 Start, preview, pause and stop

Start registers the variables with one recalculation, prepares the (correlated) random numbers and runs the iterations. The **preview window** shows the live histogram of the variable in the selected cell, or of a randomly chosen output when no variable is selected; the drop-down switches to any input or output. Below the chart are the progress, the speed, the remaining time and the running mean, standard deviation, P5 and P95.

- **Pause** holds the simulation; the bar turns yellow and **Resume** continues it.
- **Stop** ends the run early. The completed iterations become the result (the summary says “Simulation stopped after N of M iterations”); a multiple simulation skips the remaining variants. In Excel, **Esc** also stops.
- After the run, a summary lists the mean, standard deviation and the worst, real and best case of every output, and the result window opens for the selected variable.
- **Reset** forgets the results.

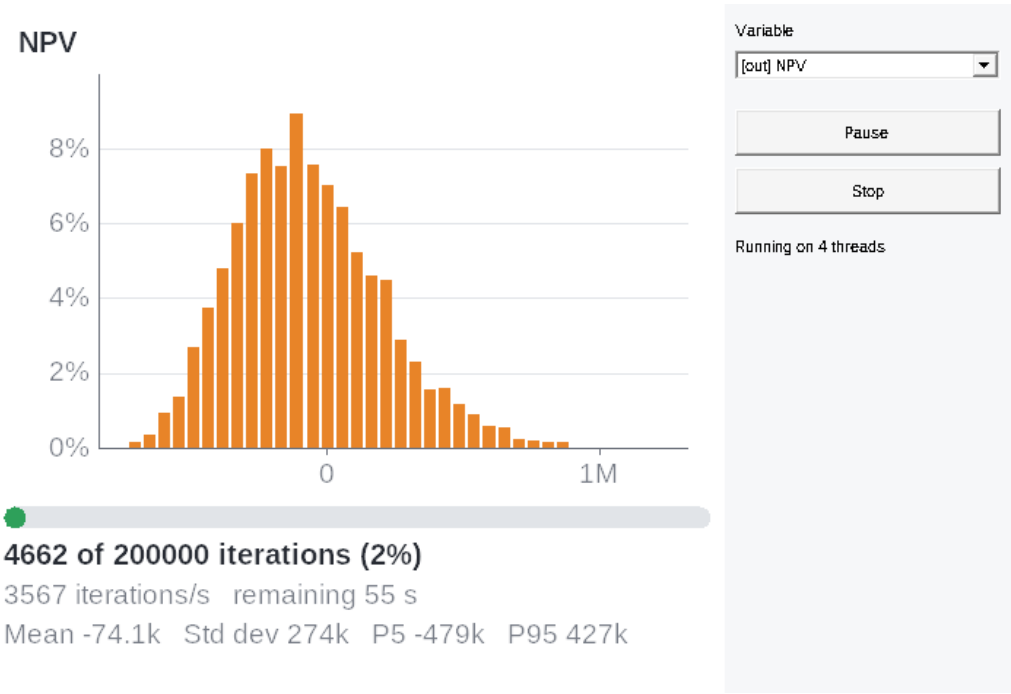


Figure 4 – Preview window during a run on four threads

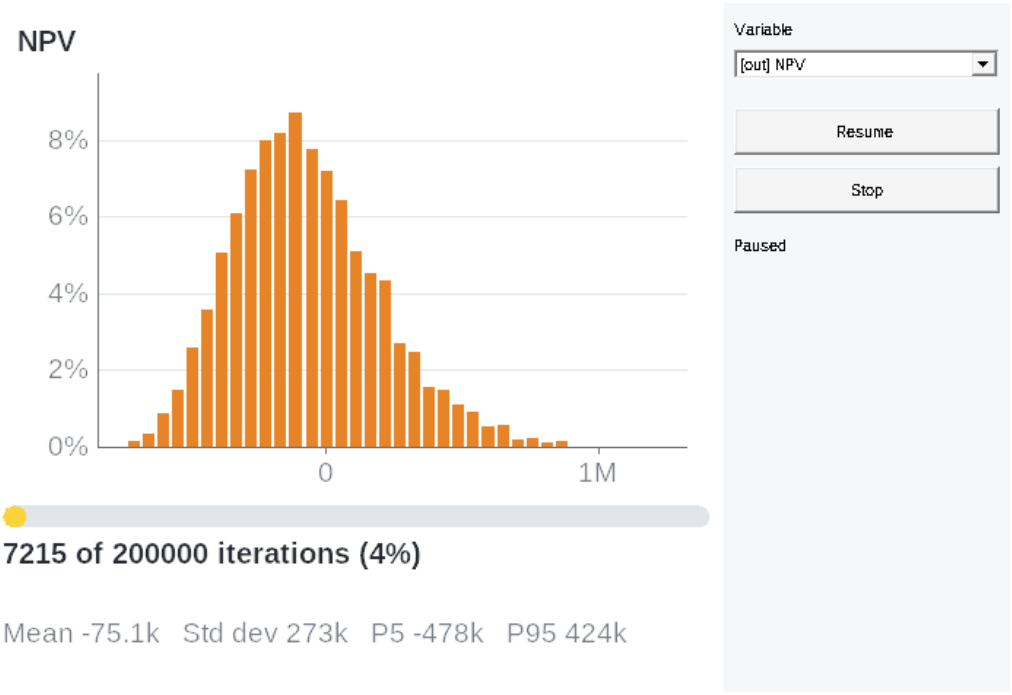


Figure 5 – The same run paused: the bar turns yellow, the button reads Resume

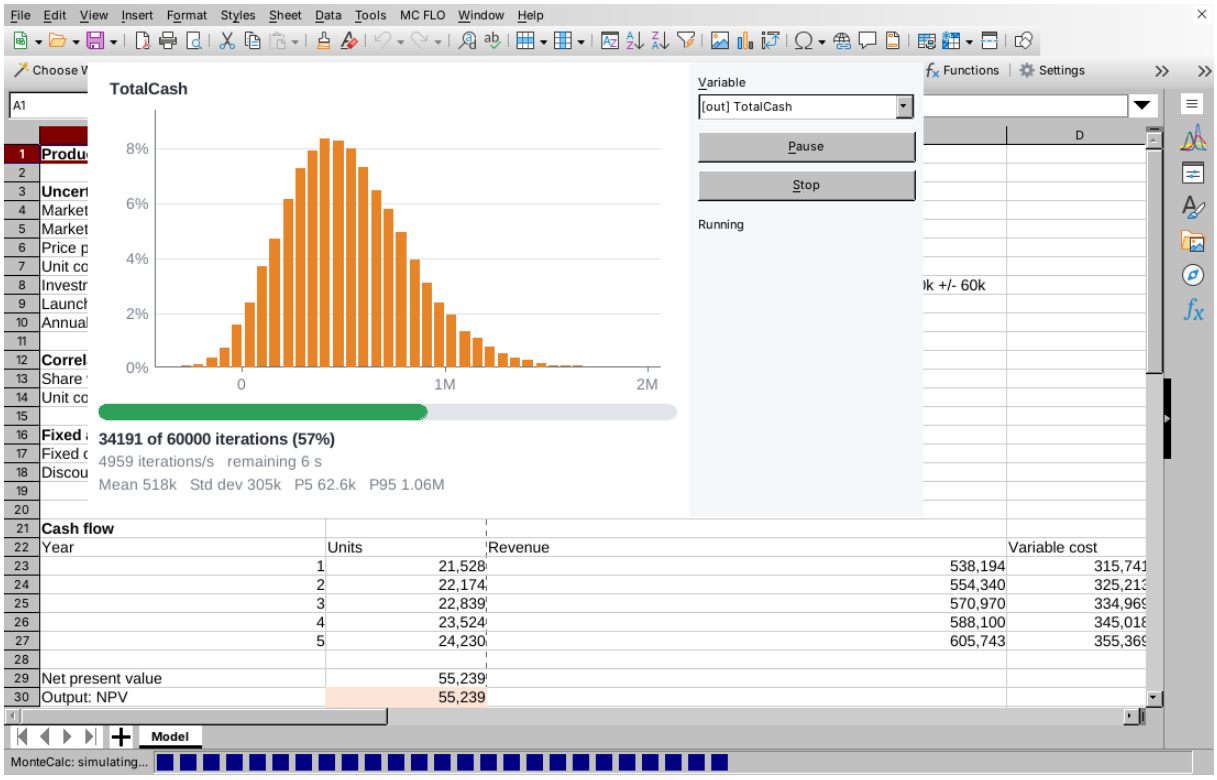


Figure 6 – The preview window in LibreOffice Calc, in front of the example model

6.3 Memory Calculation and multiple cores

Normally every iteration recalculates the spreadsheet. LibreOffice and Excel allow that only on their main thread, so one processor core does the work. With **Memory Calculation**, MonteCalc reads the formulas of the model once and evaluates them with its own engine, which is several times faster and runs the iterations on **all processor cores**. The results do not depend on the number of cores: the same seed gives the same iterations.

- If the model uses a function the engine does not know, MonteCalc says so and recalculates the spreadsheet instead.
- Models whose time series continue from one iteration to the next run in order on one core.
- In Excel the workbook is busy during a run; only the MonteCalc windows respond.

6.4 Seeds and reproducibility

With a seed other than 0, every run repeats the same iterations, so results can be compared after a change of the model. Seed −1 turns on **preview mode**: every recalculation shows new random values, a quick way to watch the model move.

7 Analysing the results

7.1 The result window

After a simulation, select any cell that holds an input, output or correlation variable: the result window shows it at once, as in MC FLO. One window follows the selection; its caption names the cell (e.g. `B30 (NPV)`), and on Windows a translucent red wedge connects the framed cell with the window, so you always see where the window belongs. The window opens beside the cell, the keyboard stays in the sheet, and the wedge follows when you move the window or scroll. A correlation cell shows the scatter of its two variables. Selecting a cell without a variable leaves the window as it is. Turn this off in Settings (Show the result window when a variable cell is selected).

You can also open it with **Result window** (Statistics group) or from the variable list. Choose the variable at the top (`[out]`, `[in]` and `[corr]` entries) and the view in the View list:

View	Shows
Histogram, Cumulative	Distribution with the magic stripe (drag it, or enter `x` or `;p`), worst and best case markers and the certainty level (e.g. 95%) with its lower and upper limit.
Tornado chart	For each input the mean of the output when the input lies in its lowest 10% (blue) and highest 10% (orange), sorted by swing. Click a bar for the scatter of that input.
Correlation (Spearman)	Rank correlation of every input and output with the variable.
Dynamic sensitivity	Range of the output by interval of an input, the expected value in orange.
Spider diagram	Pearson correlation by 10% quantile, isolated or cumulative.
Scatter / regression	The variable against its strongest input driver, or against the input you clicked. For a correlation: the scatter of its two variables with the rank correlation reached.
Variants, Box-Whisker	The variants of a multiple simulation overlaid; boxes 25–75%, whiskers 15–85%.
Decision matrix	For a target (`>` the result should exceed it, `<` stay below): the conditional probability that a driver breaches its ambition level when the target is missed, against the conditional damage. The zones blend from green (lower left) to red (upper right).
Statistics	Mean, variance, skewness, kurtosis, minimum, maximum, mode, percentiles P1…P99 and more.
Overview	Several variables at a glance: select their cells before opening the result window.

The buttons store the results in a result workbook, export the chart as an image, store it as a cell comment, start a new run and zoom (also Ctrl +, Ctrl −, Ctrl 0). All windows can be resized; they reopen with their last size.

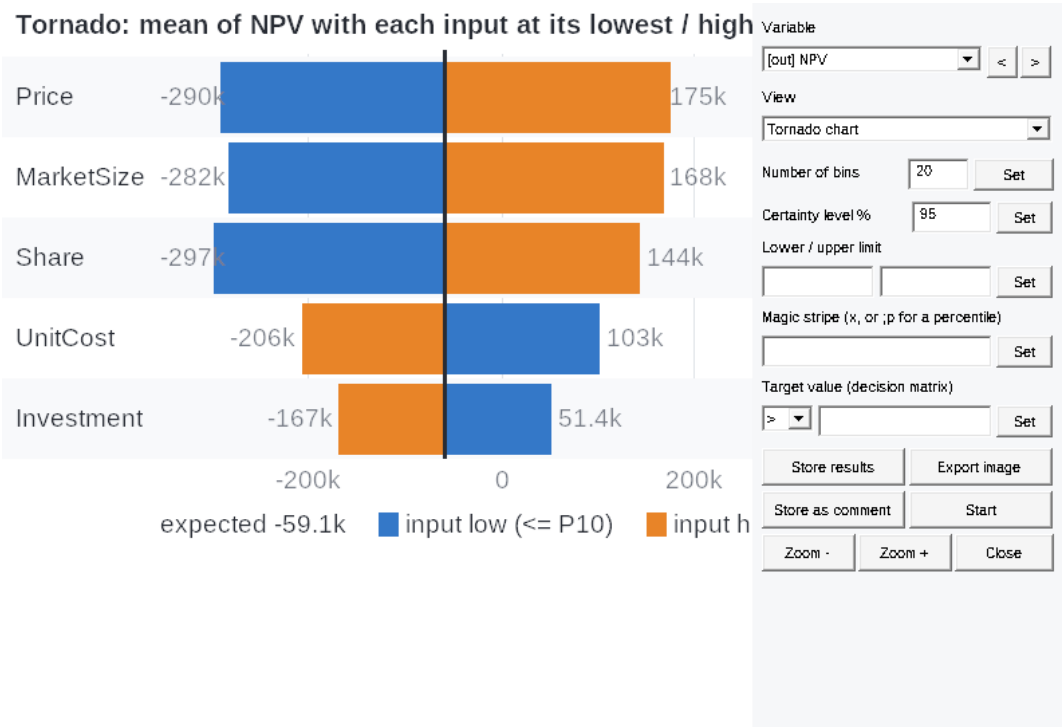


Figure 7 – Tornado chart: the NPV with each input at its lowest and highest 10%

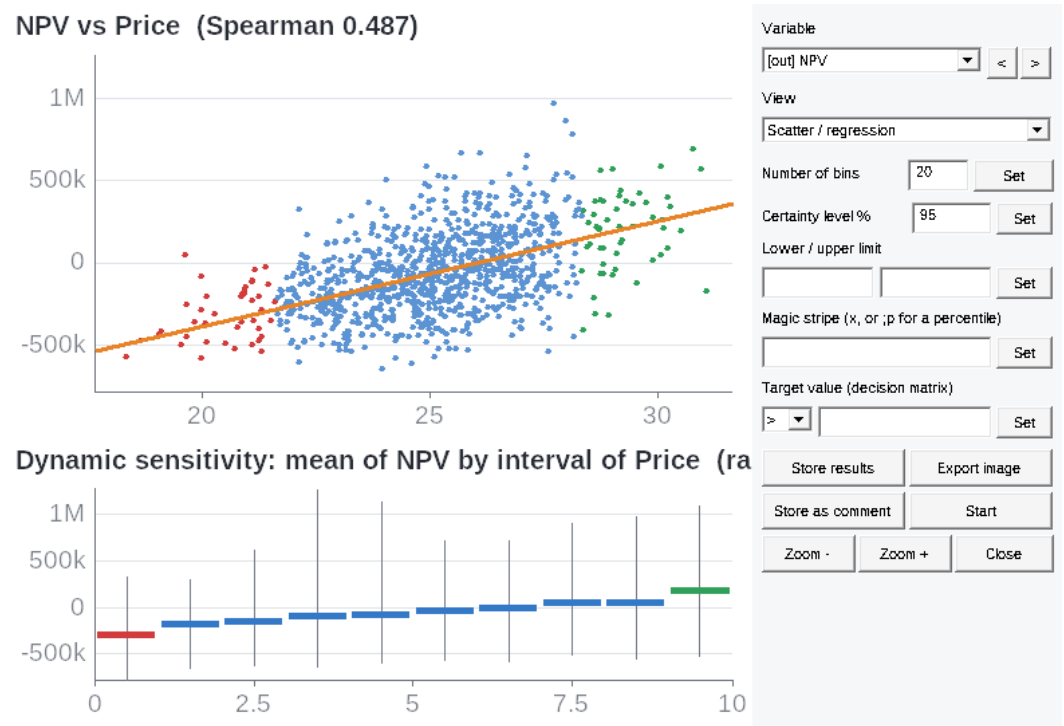


Figure 8 – Scatter / regression of the NPV against Price, with the dynamic sensitivity below

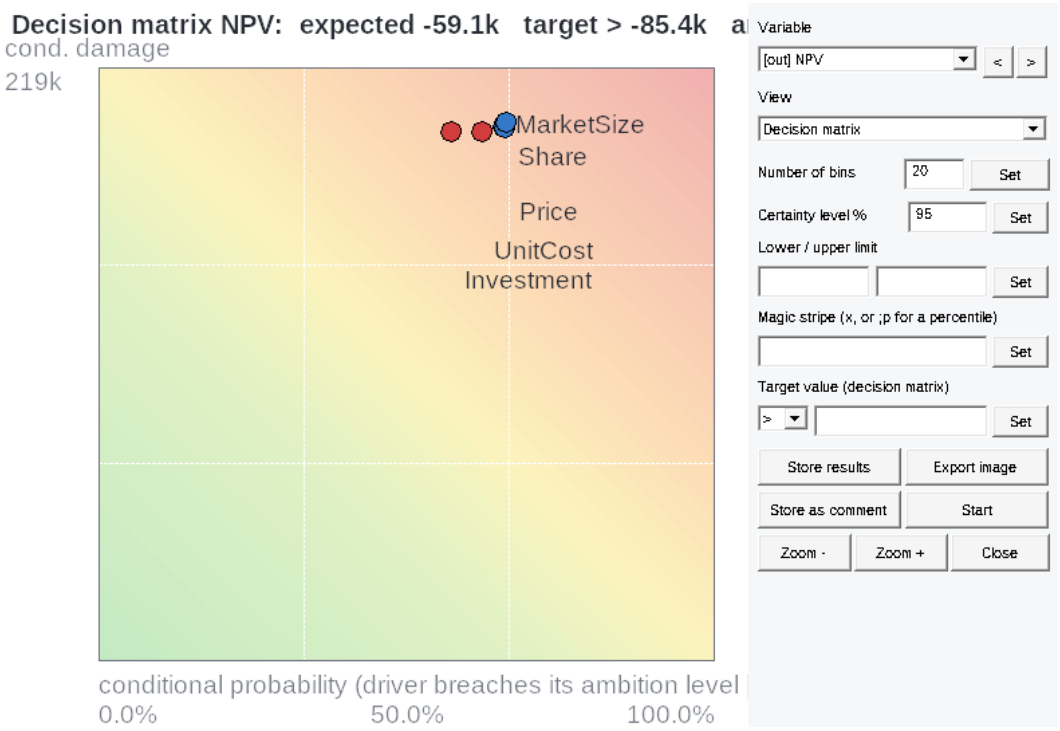


Figure 9 – Decision matrix: which drivers threaten the target, and how much

Statistic	Value	Statistic	Value
Variable	NPV	Range	2.05M
Definition	Output	Std. deviation	264k
Cell	Model!B30	Coeff. of var...	-4.463
Observations	3000	P1	-573k
Bins	20	P5	-446k
Mean	-59.1k	P10	-376k
Variance	69.5G	P25	-243k
Skewness	0.6075	P50	-85.4k
Kurtosis	0.7296	P75	100k
Minimum	-790k	P90	284k
Maximum	1.26M	P95	405k
Mode	-125k	P99	653k

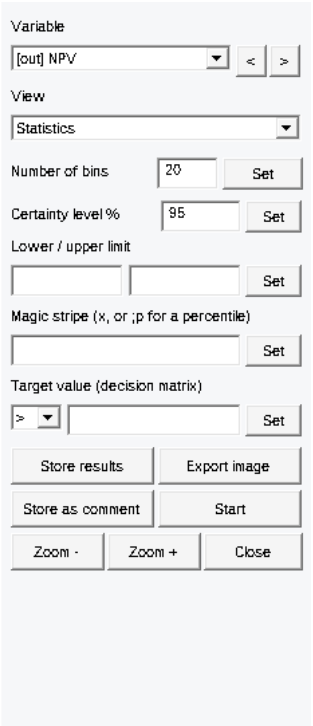


Figure 10 – Statistics view

7.2 Result functions in your cells

The `fMC_` functions bring results into the spreadsheet, so reports update with every run. A few examples:

Formula	Result
`=fMC_Average("NPV")`	mean of the NPV
`=fMC_Percentile("NPV";fMC_Worst())`	the NPV at the worst-case level (e.g. 5%)

Formula	Result
`=1-fMC_PercentileValue("NPV";0)`	probability that the NPV is positive
`=fMC_ExpectedShortfall("NPV";0.05)`	mean of the worst 5% of the NPV
`=fMC_CorrRankN("NPV";1)`	name of the input with the strongest rank correlation
`=fMC_Bayes("NPV";">";0;"Price";">=";25)`	P(NPV > 0 given Price ≥ 25)
`=fMC_Average("NPV";"High")`	mean of the NPV in variant High

The **Functions** menu (Information group) lists every `fMC_` function by category and writes the chosen one into the selected empty cell; the Function Wizard (`fx`) shows a description of each function and argument. Appendix A lists all functions. Functions that return a table spill automatically in Microsoft 365 and Excel 2021 and in LibreOffice when entered as array formula (Ctrl+Shift+Enter); in older Excel versions select the output range and confirm with Ctrl+Shift+Enter.

7.3 Analysing a single iteration

Enter an iteration number in **Choose iteration** and click **Analyze iteration** (Excel: Replay): the input values of that iteration are written into the model, so you can follow how the result came about. **Expected values** restores the formulas.

7.4 The result workbook

Store results writes a result workbook with the sheets Data (all iterations, write-protected), Summary (scenarios, 5% steps, 1% extremes), Statistics (histogram, cumulative curve, confidence interval of the mean), Sensitivity (tornado) and Variants. **Change histogram** sets the number of bins; **Close and save results** saves it as `<model>_dd_mm_yyyy_hh_mm_ss_MC_FLO` (.ods in LibreOffice, .xlsx in Excel). **Calc correlations** adds a regression sheet for an X- and Y-variable: enter a new X in I4 to get the estimate in I5 and the 95% forecast interval in K5/L5.

7.5 Influence diagram

Influence diagram (Information group) draws the model: inputs on the left, relationships in the middle, outputs on the right, with arrows traced through intermediate cells. Hover for details, drag nodes, double-click to jump to the cell. The context menu adds distributions inside the nodes, sensitivity-weighted arrows, focus on a selection, a group view and orthogonal arrows; the Results tab overlays histograms. Export as PNG, JPEG or BMP.

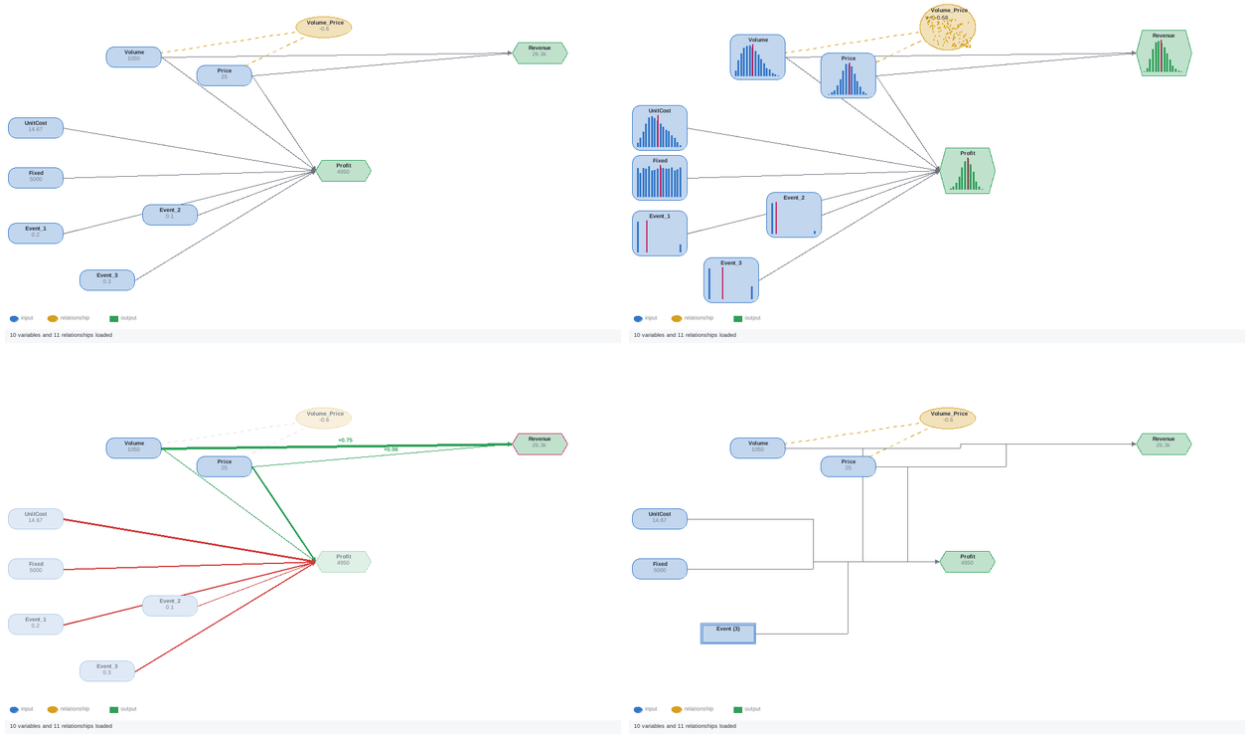


Figure 11 – Influence diagram: plain, with distributions, with sensitivity-weighted arrows and as group view

7.6 AI mentor

AI mentor opens a chat about your model. It knows the structure of the model and the statistics of the last run (not the cell contents) and can look up MC FLO function syntax. In Settings choose the answer language and add your own instructions. The conversation can be copied or exported to HTML, Word or PDF.

8 Interface reference

8.1 Excel: the MC FLO ribbon tab

Group	Commands
Model specification	Continuous, discrete and time-series distribution menus (each opens the wizard; on a cell that holds a variable it opens that variable), Wizard, Insert correlation, Insert output, Register, Lock, Fit distribution
Information	Influence diagram, Functions (the `fMC_` functions by category, plus the Function Wizard), Model, Show variables, AI mentor
Simulation	Best / Real / Worst levels and Iterations, check boxes Use correlations, Visualize, Memory Calculation and Preview window, Multiple simulation, Start, Reset, Settings
Statistics	Variable drop-down, Show statistical parameters, Result window, Select variant
Correlations	X / Y drop-downs, Calc correlations, Correlogram
Iteration	Iteration number, Replay, Expected values
Results	Close and save results, Store results, Change histogram
Other	Sort ascending / descending, Model check, Forecast tool, Batch, Hide / load formulas, Variables, FLO_Context sheet, Preview (random values), Language, About MonteCalc (also the large MC FLO logo button)

The commands are also macros (`MCFLO.Start`, `MCFLO.Reset`, `MCFLO.Settings`, ...) that VBA can call with `Application.Run "MCFLO.Start"`.

8.2 LibreOffice: sidebar, menu, toolbar and tab

LibreOffice extensions cannot add a ribbon tab of their own, so the same groups appear in four places that run the same commands:

- the **MC FLO sidebar deck** (icon at the right edge), one panel per group, with the input fields of the Simulation group;
- the **MC FLO menu**, whose submenus follow the groups and include the distribution pickers with a small picture of every distribution;
- the **MC FLO toolbar**;
- the **Extension** tab of the tabbed interface (View → User Interface → Tabbed).

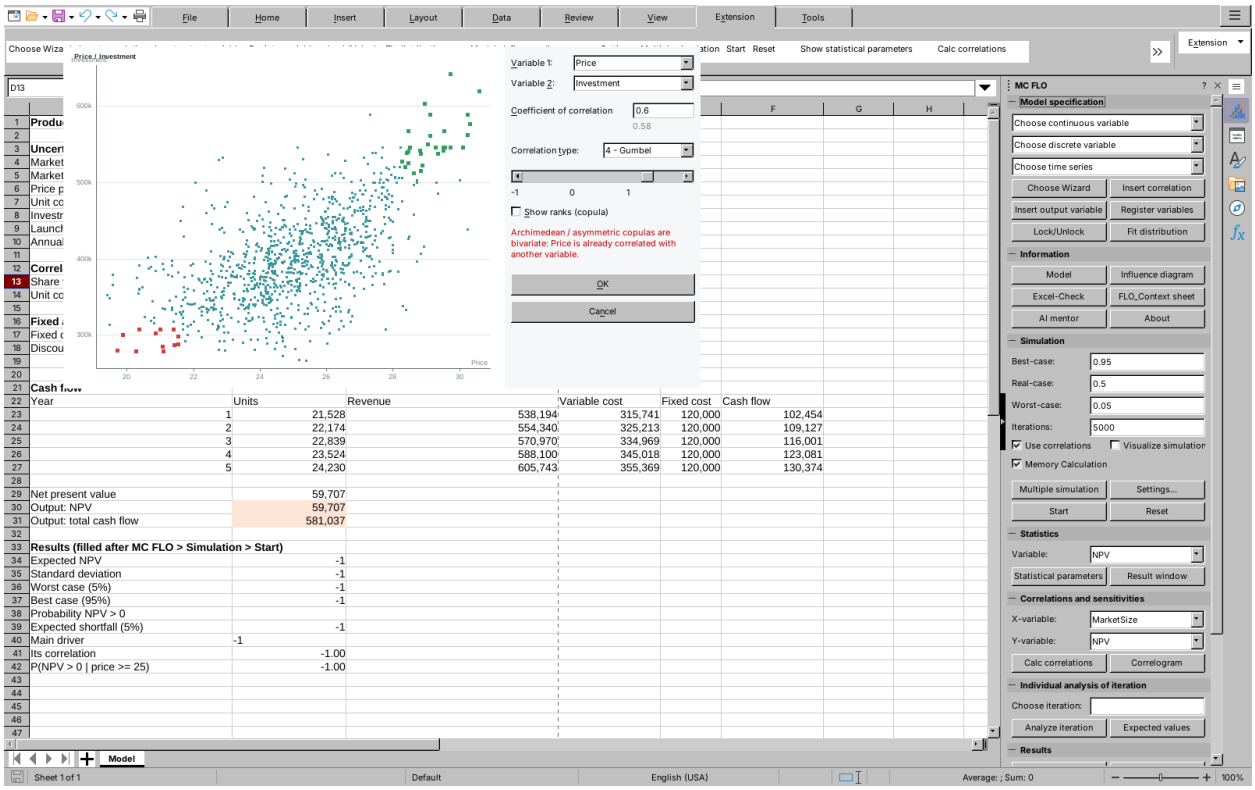


Figure 12 – LibreOffice Calc with the MC FLO sidebar deck (right), the Extension tab (top) and the correlation window

8.3 Keyboard

Keys	Where	Action
Ctrl +, Ctrl −, Ctrl 0	all chart windows	zoom in, zoom out, reset
Tab	all windows	next field
Esc	Excel, during a run	stop the simulation (keeps the completed iterations)
F5, Ctrl+R/L/S/E/F/T/G/D/I/H/K/J, Ctrl+M	influence diagram	the commands of its context menu

9 Differences between the editions

Topic	LibreOffice	Excel
Commands	sidebar deck, menu, toolbar, Extension tab	MC FLO ribbon tab (Add-ins tab as fallback)
Formula separator	`;`	`,` (English) or `;` (German, Spanish); MonteCalc handles both
Result workbook	.ods	.xlsx
Function descriptions	in the Function Wizard, in the chosen language	in the Function Wizard, in English (categories in the chosen language)
During a run	LibreOffice keeps redrawing; the preview window responds	Excel is busy; only the MonteCalc windows respond

Everything else – functions, windows, settings, results – is the same code in both editions.

10 Troubleshooting

LibreOffice shows #NAME? for every MonteCalc function.

LibreOffice has not loaded the extension. Restart LibreOffice completely (on Windows including the Quickstarter in the system tray). If it persists, check in Tools → Extension Manager that MonteCalc is installed and enabled, remove and add it again, and make sure you installed the file for your system (Windows or Linux, 64-bit).

Excel does not show the MC FLO tab.

Enable **MonteCalc** under File → Options → Add-ins → Manage: COM Add-ins. All commands are also on the **Add-ins** tab. If Excel refuses to load the .xll, unblock the file (Properties → Unblock) and check that you use the 64-bit file for 64-bit Office and the 32-bit file for 32-bit Office.

A formula entered with commas shows the wrong values (German or Spanish Excel).

Excel in German or Spanish uses `;` between arguments. `=FLOsimula_PERT(0,5;1;0,8;"Share")` is correct there; in the MonteCalc windows you may type decimal commas, they are converted.

A function returns -1 or an error after opening the workbook.

Result functions (`fMC_…`) need a simulation and return −1 until one has run. Run **Start**.

The simulation is slow.

Turn on **Memory Calculation**. Turn off Visualize simulation. Reduce the iterations for a first look.

Memory Calculation reports that a function is not supported.

The model uses a spreadsheet function that MonteCalc's engine does not evaluate; the run recalculates the spreadsheet instead and the results are correct, only slower.

The result differs from the last run.

Seed 0 draws new random numbers every run. Set a seed (e.g. 2026) in Settings to repeat the same iterations.

A correlation is refused.

The pair is already correlated in another cell, or one of the inputs is already part of an Archimedean or asymmetric correlation (these are bivariate). The correlation window explains which.

Saving from the wizard or the correlation window fails in Excel.

Make sure no cell is being edited in Excel (press Esc in the sheet) and try again. The error message shows the formula Excel refused and Excel's own description; please include it when reporting a problem.

Appendix A Function reference

All 178 functions of MonteCalc, generated from the table that also generates the add-ins. Arguments in [brackets] are optional. Separate arguments with ; (, in English Excel).

Model functions: continuous distributions

Function	Description
`FLOsimula_Uniform(Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Continuous uniform distribution.
`FLOsimula_Uniforme(Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Continuous uniform distribution. Alias of FLOsimula_Uniform (MC FLO compatibility).
`FLOsimula_Triangular(Lower_Limit; Upper_Limit; Mode; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Triangular distribution.
`FLOsimula_PERT(Lower_Limit; Upper_Limit; Mode; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: PERT distribution.
`FLOsimula_Normal(Mean; StDeviation; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Normal distribution.
`FLOsimula_LogNormal(Shift; Mean; StDeviation; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Log-normal distribution: Shift + exp(Normal(Mean; StDeviation)).
`FLOsimula_Beta(Param_alfa; Param_beta; Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Beta distribution between Lower_Limit and Upper_Limit.
`FLOsimula_Gumbel(Shift; Param_alfa; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Gumbel (extreme value) distribution: location Shift, scale param_alfa.
`FLOsimula_Gamma(Shift; Param_alfa; Param_beta; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Gamma distribution: Shift + Gamma(scale param_alfa, shape param_beta).
`FLOsimula_Exponential(Shift; Param_alfa; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Exponential distribution: Shift + Exponential(mean param_alfa).
`FLOsimula_TruncNormal(Mean; StDeviation; Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Truncated normal distribution.
`FLOsimula_TruncatedNormal(Mean; StDeviation; Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Truncated normal distribution. Alias of FLOsimula_TruncNormal (MC FLO compatibility).
`FLOsimula_Weibull(Shift; Param_alfa; Param_beta; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Weibull distribution: Shift + Weibull(scale param_alfa, shape param_beta).
`FLOsimula_chi2(Freedom; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Chi-square distribution.
`FLOsimula_studentt(Freedom; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Student-t distribution.
`FLOsimula_Pareto(Shift; Param_alfa; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Pareto distribution with minimum Shift (> 0) and shape param_alfa.
`FLOsimula_Fisher(Freedom1; Freedom2; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: F distribution.
`FLOsimula_ArcSin(Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Arcsine distribution.
`FLOsimula_Cauchy(Shift; Param_alfa; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Cauchy distribution: location Shift, scale param_alfa (the cell shows the median).
`FLOsimula_Frechet(Start; Shift; Param_alfa; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Frechet distribution: minimum Start, scale Shift, shape param_alfa.
`FLOsimula_Logistic(Location; Shift; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Logistic distribution: location Location, scale Shift.

Function	Description
`FLOsimula_UserDefined(Data; NumberOfBins; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: user-defined distribution from a data area. Two-column areas are read as (quantile, value) rows; if the quantiles are not all in [0,1] the last row is (minimum, maximum), as in the manual.

Model functions: discrete distributions

Function	Description
`FLOsimula_DiscreteUniform(Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Discrete uniform distribution (integers).
`FLOsimula_DiscreteUniforme(Lower_Limit; Upper_Limit; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Discrete uniform distribution (integers). Alias of FLOsimula_DiscreteUniform (MC FLO compatibility).
`FLOsimula_Bernoulli(Probability; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Bernoulli distribution: 1 (TRUE) with the given probability, else 0.
`FLOsimula_Binomial(Probability; Tries; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Binomial distribution: successes in Tries trials.
`FLOsimula_Poisson(Occurency; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Poisson distribution.
`FLOsimula_Pascal(Probability; Success; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Pascal (negative binomial) distribution: failed attempts until Success successes; Probability = probability of a failure.
`FLOsimula_Geometric(Probability; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Geometric distribution: failures before the first success.
`FLOsimula_Hypergeometric(Tries; Population; Success_Possibles; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Hypergeometric distribution (draws without replacement).
`FLOsimula_DiscreteUserDefined(Values; Probabilities; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: user-defined discrete distribution.

Model functions: time series

Function	Description
`FLOsimula_TemporalS_ARmodel(Start; Weight; StDeviation; UseBefore; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: AR(1) process around the level Start. The (correlated) random number drives the noise of each iteration; UseBefore = 1 restarts every iteration at Start.
`FLOsimula_TemporalS_MAmodel(Start; Weight; StDeviation; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: MA(1) process. The (correlated) random number drives the noise of each iteration; UseBefore = 1 restarts every iteration at Start.
`FLOsimula_TemporalS_ARMAmodel(Start; Weight1; StDeviation1; Weight2; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: ARMA(1,1) process. The (correlated) random number drives the noise of each iteration; UseBefore = 1 restarts every iteration at Start.
`FLOsimula_TemporalS_ARCHmodel(Omega; Alpha; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: ARCH(1) process (mean 0, non-constant volatility). The (correlated) random number drives the noise of each iteration; UseBefore = 1 restarts every iteration at Start.
`FLOsimula_TemporalS_WPmodel(Start; Drift; Mu; StDeviation; UseBefore; Variable_name; [Number]; [Min]; [Max]; [Round])`	Input variable: Wiener process (geometric Brownian motion); StDeviation is optional. The (correlated) random number drives the noise of each iteration; UseBefore = 1 restarts every iteration at Start.

Model functions: output, correlation, bounds

Function	Description
`FLOsimula_Bounded([Variable]; [LowerLimit]; [UpperLimit]; [Variable_name])`	Input variable: truncates an existing MC FLO variable to [LowerLimit, UpperLimit]. Only values within the limits are drawn. Before a simulation the cell shows the referenced variable's expected value.
`FLOsimula_output([Name]; Value)`	Output variable.
`FLOsimula_correlation([Variable1]; [Variable2]; [Coef_corr]; [CorType])`	Correlation between two input variables; returns the coefficient. Type codes as in the MC FLO wiki. Archimedean and asymmetric correlations are bivariate.

Result functions (need a simulation)

Function	Description
`fMC_Average(Name; [Referenz])`	Average of the distribution. Returns -1 if no simulation has been run.
`fMC_Mean(Name; [Referenz])`	Average of the distribution. Returns -1 if no simulation has been run.
`fMC_Min(Name; [Referenz])`	Minimum of the distribution. Returns -1 if no simulation has been run.
`fMC_Max(Name; [Referenz])`	Maximum of the distribution. Returns -1 if no simulation has been run.
`fMC_Sum(Name; [Referenz])`	Sum of the distribution. Returns -1 if no simulation has been run.
`fMC_StdS(Name; [Referenz])`	Sample standard deviation. Returns -1 if no simulation has been run.
`fMC_Variance(Name; [Referenz])`	Sample variance. Returns -1 if no simulation has been run.
`fMC_Skewness(Name; [Referenz])`	Skewness. Returns -1 if no simulation has been run.
`fMC_Kurtosis(Name; [Referenz])`	Excess kurtosis. Returns -1 if no simulation has been run.
`fMC_CV(Name; [Referenz])`	Coefficient of variation. Returns -1 if no simulation has been run.
`fMC_Semivariance(Name; [Referenz])`	Semivariance (below the mean). Returns -1 if no simulation has been run.
`fMC_SemivarianceL(Name; Limit; [Referenz])`	Semivariance of the values below Limit. Returns -1 if no simulation has been run.
`fMC_SemivarianceR(Name; Limit; [Referenz])`	Semivariance of the values above Limit. Returns -1 if no simulation has been run.
`fMC_ExpGain(Name; [Referenz])`	Average of all values above 0. Returns -1 if no simulation has been run.
`fMC_ExpGainV(Name; [Referenz])`	Variance of all values above 0. Returns -1 if no simulation has been run.
`fMC_ExpLoss(Name; [Referenz])`	Average of all values below 0. Returns -1 if no simulation has been run.
`fMC_ExpectedShortfall(Name; Percentile; [Referenz])`	Expected shortfall beyond the value-at-risk at the percentile (<= 0.5 left tail, else right tail). Returns -1 if no simulation has been run.
`fMC_Percentile(Name; Percentile; [Referenz])`	Value at the percentile: the ROUNDUP(n * Percentile)-th smallest value. Returns -1 if no simulation has been run.
`fMC_PercentileValue(Name; Value; [Referenz])`	Percentile (0..1) of a given value: position of the first value >= Value among the sorted values. Returns -1 if no simulation has been run.
`fMC_PercentileInterval(Name; Lower_Limit; Upper_Limit; [Referenz])`	Confidence level (share of iterations) between the limits. Returns -1 if no simulation has been run.
`fMC_Mode(Name; Number_bins; [Referenz])`	Most frequent value based on a histogram. Returns -1 if no simulation has been run.

Function	Description
`fMC_Density(Name; Value; [Number_bins]; [Referenz])`	Share of the iterations at Value (discrete) or in the histogram column containing Value. Returns -1 if no simulation has been run.
`fMC_ExtremeValue(Name; Pos; [Referenz])`	Value at position Pos of the sorted list. Returns -1 if no simulation has been run.
`fMC_PosValue(Name; Pos; [Referenz])`	Value at iteration Pos. Returns -1 if no simulation has been run.
`fMC_ConditionalPosValue(Name; Value; [Referenz])`	Position (iteration) whose value equals Value, or the nearest one. Returns -1 if no simulation has been run.
`fMC_ConditionPosValue(Name; Value; [Referenz])`	Position (iteration) whose value equals Value, or the nearest one (alias of fMC_ConditionalPosValue). Returns -1 if no simulation has been run.
`fMC_ADs(Name; [Referenz])`	Anderson-Darling normality test of a variable: 1 if very likely normal, else 0. Returns -1 if no simulation has been run.
`fMC_CorrP(Name1; Name2; [Referenz])`	Pearson correlation between two variables. Returns -1 if no simulation has been run.
`fMC_CorrS(Name1; Name2; [Referenz])`	Spearman rank correlation between two variables. Returns -1 if no simulation has been run.
`fMC_RegressionX(Name1; Name2; [Referenz])`	Slope of the linear regression (Name1 = x, Name2 = y). Returns -1 if no simulation has been run.
`fMC_RegressionY(Name1; Name2; [Referenz])`	Intercept of the linear regression (Name1 = x, Name2 = y). Returns -1 if no simulation has been run.
`fMC_Spider(Name1; Name2; Value1; Value2; [Referenz])`	Pearson correlation of the two variables within the range Value1..Value2 of Name1. Returns -1 if no simulation has been run. Value1 and Value2 are quantiles of Name1 when both lie in [0,1], otherwise values of Name1.
`fMC_DSensitivity(Name1; Name2; DSensotype; NumberBins; [Referenz])`	Dynamic sensitivity between input Name1 and output Name2: 0 range, 1 minimum, 2 maximum of the interval means. Returns -1 if no simulation has been run.
`fMC_FinancialValue(Name1; Name2; Value2; Diff; Value1; [Referenz])`	Percentile Value1 of event A (Name1) given that event B (Name2) is within Value2 +/- Diff. Returns -1 if no simulation has been run.
`fMC_Bayes(Name1; Comparison1; Value1; Name2; Comparison2; Value2; [Referenz])`	Conditional probability P(A | B), e.g. fMC_Bayes("NPV";">=";1000;"Costs";">=";100). Returns -1 if no simulation has been run.
`fMC_BayesV([Name1]; [Vergleich1]; [Wert1]; [Name2]; [Vergleich]; [Wert2]; [Referenz])`	Average of A (Name1) over the iterations where A and event B (Name2) meet their conditions. Returns -1 if no simulation has been run.
`fMC_BayesV_MC([Name1]; [Vergleich1]; [Wert1]; [Name2]; [Vergleich]; [Wert2]; [Referenz])`	Average of A (Name1) over the iterations where A and event B (Name2) meet their conditions. Returns -1 if no simulation has been run.
`fMC_Bayes4([Name1]; [Compare1]; [Value1]; [Name2]; [Compare2]; [Value2]; [Name3]; [Compare3]; [Value3]; [Name4]; [Compare4]; [Value4]; [Name5]; [Compare5]; [Value5]; [Referenz])`	P(A | B, C, D, E) for up to five events; a condition may also be given as one text (">="&100). Returns -1 if no simulation has been run.
`fMC_Bayes4V([Name1]; [Compare1]; [Value1]; [Name2]; [Compare2]; [Value2]; [Name3]; [Compare3]; [Value3]; [Name4]; [Compare4]; [Value4]; [Name5]; [Compare5]; [Value5]; [Referenz])`	Average of A over the iterations where A and the events B..E meet their conditions. Returns -1 if no simulation has been run.
`fMC_Bayes4V_MC([Name1]; [Compare1]; [Value1]; [Name2]; [Compare2]; [Value2]; [Name3]; [Compare3]; [Value3]; [Name4]; [Compare4]; [Value4]; [Name5]; [Compare5]; [Value5]; [Referenz])`	Average of A over the iterations where A and the events B..E meet their conditions. Returns -1 if no simulation has been run.
`fMC_CorrRankN(Name; Rank; [Referenz])`	Name of the input with the Rank-th highest (absolute, Pearson) correlation with the output.

Function	Description
fMC_CorrRank(Name; Rank; [Referenz])	Pearson correlation of the Rank-th strongest input with the output. Returns -1 if no simulation has been run.
fMC_AD([NameOrData])	Anderson-Darling normality test on a variable name or a data area: TRUE (1) if very likely normal.
fMC_BootStrapN(Name; NumNewExamples; Confidence; [Percentile])	Bootstrap confidence interval of the mean (or of Percentile) of a variable: "mean; lower; upper".
fMC_Kernel(Name; AdaptRate; Number_bins; [Bandwith]; [ReturnHistogram])	Kernel density estimate of a variable: n x 1 smoothed sample or (ReturnHistogram) bins x 3 histogram (lower, upper, density).
fMC_PredictiveRegressionLine(Name1; Name2; Value; [Percentile])	Typical value at Value from simulated slope (Name1) and intercept (Name2); Percentile 0 = expected value.
fMC_PredictiveRegressionLineValue(Name1; Name2; Name3; Value; [Percentile])	Predictive value at Value from simulated slope, intercept and standard error of regression.

Information and settings

Function	Description
fMC_Iterations()	Number of iterations of the last simulation.
fMC_Best(Value)	Confidence level of the best case; with Value in [0,1] the best case is set dynamically to Value.
fMC_Real(Value)	Confidence level of the real case; with Value in [0,1] the real case is set dynamically to Value.
fMC_Worst(Value)	Confidence level of the worst case; with Value in [0,1] the worst case is set dynamically to Value.
fMC_GetInputName(Pos)	Input variable at position Pos.
fMC_GetUserDefinedInputName(Pos)	User-defined input variable at position Pos.
fMC_GetOuputName(Pos)	Output variable at position Pos (name as spelled in the manual).
fMC_GetOutputName(Pos)	Output variable at position Pos.
fmc_Name(Name)	Name of the MC FLO variable in the given cell.
fMC_Bin(Value)	Sets the number of histogram columns used by all histograms and functions (global).
fMC_SimulaOn(Value)	Workbook simulation mode: TRUE shows random values after each recalculation, FALSE the expected values.

Data tools

Function	Description
fMC_Pearson(column1; column2)	Pearson correlation of two data rows/columns.
fMC_PearsonV(column1; column2)	Volatile variant of fMC_Pearson.
fMC_Spear(column1; column2)	Spearman rank correlation of two data rows/columns.
fMC_Sort(column1; pos; [Sort])	Value at position pos of the sorted list (Sort 0 ascending, 1 descending).
fMC_FitC(column1; column2)	Best correlation (copula) type 0..8 for two data series.
fMC_BSMVal(Basis; Time; Interest; Sigma)	Price of the underlying after Time periods (Black-Scholes-Merton, risk-neutral): random in a simulation, expected value otherwise.
fMC_BootStrap(Matrix1; NumNewExamples; Confidence; [Percentile])	Bootstrap confidence interval of the mean (or Percentile) of a data area: "mean; lower; upper".
fMC_BootStrapVector(Matrix1; NumNewExamples; [Percentile]; [ReturnPercentH]; [Compound])	Bootstrap distribution of the mean (or Percentile / hit rate / compound sum) of a data area, n x 1.

Function	Description
fMC_Kernel2(Data; [LowerLimit]; [UpperLimit]; [Bandwith]; [Number_bins]; [KernelType]; [ReturnHistogram]; [ReturnSample]; [Sample])	Kernel density estimate of a data area over bins: bins x 1 densities or bins x 3 (lower, upper, density).
fMC_AD_Data(Matrix1)	Anderson-Darling normality test on a data area (same as fMC_AD with a range).
fMC_FinancialDTrans(Initial; Periods; ActualYear; Amount)	Share of Amount in ActualYear when it is spread evenly over Periods years starting at Initial.

Regression

Function	Description
fMC_BootStrapVectorLinearRegressionX(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression])	Bootstrap distribution of the slope of a simple linear regression, n x 1.
fMC_BootStrapVectorLinearRegressionY(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression])	Bootstrap distribution of the intercept of a simple linear regression, n x 1.
fMC_BootStrapVectorLinearRegressionRE(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression])	Bootstrap distribution of the standard error of regression of a simple linear regression, n x 1.
fMC_BootStrapVectorMLRegression(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression])	Bootstrap samples of a multiple linear regression: rows (intercept, coefficients..., standard error).
fMC_BootStrapVectorMLRegressionParameters(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression]; [CI]; [FeatureDescription])	Parameter summary of a bootstrapped multiple regression: (m+2) x 4 (label, mean, lower, upper).
fMC_BayesPredictionRegression(DataY; DataX; Value; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Forecast of a Bayesian linear regression at Value (mean, or Percentile of the predictive distribution). Posterior from the Bayesian bootstrap (Dirichlet-weighted least squares).
fMC_BayesPredictionRegressionParameters(DataY; DataX; [CI]; [FeatureDescription]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Parameters of a Bayesian linear regression: (m+2) x 4 (label, mean, lower, upper).
fMC_BayesPredictionRegressionCIP(TrainDataY; TrainDataX; TestDataY; TestDataX; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Coverage rate of the test data by the central predictive interval (1 = as expected).
fMC_BayesPredictionRegressionMAE(TrainDataY; TrainDataX; TestDataY; TestDataX; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Mean absolute error of the Bayesian regression on the test data.
fMC_BayesPredictionRegressionResiduals(TrainDataY; TrainDataX; TestDataY; TestDataX; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Residuals of the test data under the Bayesian regression, n x 1.
fMC_BayesPredictionRegressionV_MC(DataY; DataX; Value; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Forecast of a Bayesian linear regression at Value (mean, or Percentile of the predictive distribution). Same as fMC_BayesPredictionRegression (MC FLO Memory Calculation name).
fMC_BayesPredictionRegressionCIPV_MC(TrainDataY; TrainDataX; TestDataY; TestDataX; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Coverage rate of the test data by the central predictive interval (1 = as expected). Same as fMC_BayesPredictionRegressionCIP (MC FLO Memory Calculation name).
fMC_BayesPredictionRegressionMAEV_MC(TrainDataY; TrainDataX; TestDataY; TestDataX; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])	Mean absolute error of the Bayesian regression on the test data. Same as fMC_BayesPredictionRegressionMAE (MC FLO Memory Calculation name).

Function	Description
`fMC_BayesPredictionRegressionParameters_MC(DataY; DataX; [CI]; [FeatureDescription]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])`	Parameters of a Bayesian linear regression: (m+2) x 4 (label, mean, lower, upper). Same as fMC_BayesPredictionRegressionParameters (MC FLO Memory Calculation name).
`fMC_BayesPredictionRegressionResiduals_MC(TrainDataY; TrainDataX; TestDataY; TestDataX; [Percentile]; [UseLogForY]; [UseRepeatedMedianRegression]; [NumNewSample]; [kfolds])`	Residuals of the test data under the Bayesian regression, n x 1. Same as fMC_BayesPredictionRegressionResiduals (MC FLO Memory Calculation name).
`fMC_BootStrapVectorMLRegression_MC(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression])`	Bootstrap samples of a multiple linear regression: rows (intercept, coefficients..., standard error). Same as fMC_BootStrapVectorMLRegression (MC FLO Memory Calculation name).
`fMC_BootStrapVectorMLRegressionParameters_MC(DataY; DataX; NumberNewSamples; [UseRepeatedMedianRegression]; [CI]; [FeatureDescription])`	Parameter summary of a bootstrapped multiple regression: (m+2) x 4 (label, mean, lower, upper). Same as fMC_BootStrapVectorMLRegressionParameters (MC FLO Memory Calculation name).

Bayesian statistics

Function	Description
`fMC_BayesMCMC(Name1; Name2; BurnIn; Thinning; NumberIterations; [ReturnData]; [Number_bins]; [Number_binsDivide]; [Referenz])`	Posterior of prior Name1 and likelihood Name2 by Markov chain Monte Carlo (Gaussian proposal): n x 1 data or histogram. The posterior density is proportional to the product of the two (kernel-estimated) densities.
`fMC_BayesMCMCA(Name1; Name2; BurnIn; Thinning; NumberIterations; [ReturnData]; [Number_bins]; [Number_binsDivide]; [Referenz])`	Posterior of prior Name1 and likelihood Name2 by Markov chain Monte Carlo (asymmetric uniform proposal): n x 1 data or histogram. The posterior density is proportional to the product of the two (kernel-estimated) densities.
`fMC_BayesMCMCV(Name1; Name2; BurnIn; Thinning; NumberIterations; [ReturnData]; [Number_bins]; [Number_binsDivide]; [Referenz])`	Posterior of prior Name1 and likelihood Name2 by Markov chain Monte Carlo (volatile variant): n x 1 data or histogram. The posterior density is proportional to the product of the two (kernel-estimated) densities.
`fMC_BayesMCMCAV(Name1; Name2; BurnIn; Thinning; NumberIterations; [ReturnData]; [Number_bins]; [Number_binsDivide]; [Referenz])`	Posterior of prior Name1 and likelihood Name2 by Markov chain Monte Carlo (volatile variant): n x 1 data or histogram. The posterior density is proportional to the product of the two (kernel-estimated) densities.
`fMC_BayesPrediction(Name1; [Name2]; Count; NumberIterations; Attempts; [ReturnData]; [Number_bins]; [Referenz])`	Predictive distribution of the failures before Count successes, using posterior success probabilities. The manual leaves the predictive model open; MonteCalc uses a negative-binomial predictive per posterior draw.
`fMC_BayesCombine([Name1]; [Share1]; [Name2]; [Share2]; [Name3]; [Share3]; [Name4]; [Share4]; [Name5]; [Share5]; [ReturnData]; [NumBin]; [Referenz])`	Combines up to five prior distributions into one (mixture): n x 1 data or histogram.
`fMC_BayesMCMCV_Robust(Vector1; Vector2)`	Posterior of two data vectors (prior, likelihood) by a grid approximation (robust for sparse data), n x 1.
`fMC_BayesClassifier(FeatureData; ClassData; PredictData; [Testproportion]; [ModelSelection]; [ReturnConfusionMatrixTestSet]; [ReturnPredictionInterval]; [PredictionInterval]; [Iterations]; [HisBins]; [LowerLimit]; [UpperLimit]; [FeatureImpact]; [FeatureDescription]; [kfolds]; [UseFlatPrior]; [UseWeightedFeatures])`	Naive Bayes classification (predictions, confusion matrix, credible intervals or feature impact). Feature impact is the drop in test accuracy when the feature is permuted.

Function	Description
`fMC_BayesFactor(ExpectedProbability; ObservedProbability; NumberOfAttempts; [NumNewSample]; [Percentile]; [Attempts])`	Bayes factor of H1 (p unknown) against H0 (p = ExpectedProbability) in a binomial model; with Percentile, the posterior credible value. Computed analytically (uniform prior under H1), so NumNewSample and Attempts are not needed.
`fMC_BayesClassifier_MC(FeatureData; ClassData; PredictData; [Testproportion]; [ModelSelection]; [ReturnConfusionMatrixTestSet]; [ReturnPredictionInterval]; [PredictionInterval]; [Iterations]; [HisBins]; [LowerLimit]; [UpperLimit]; [FeatureImpact]; [FeatureDescription]; [kfolds]; [UseFlatPrior]; [UseWeightedFeatures])`	Naive Bayes classification (predictions, confusion matrix, credible intervals or feature impact). Same as fMC_BayesClassifier (MC FLO Memory Calculation name).

Batch forecasting

Function	Description
`fMC_Batch_Time(Matrix1; Number; [ResultOrder]; [Selection]; [NumIterations]; [Percentil])`	Marks a series for a batch forecast with stochastic time-series processes (run 'Start Batch').
`fMC_Batch_HW_D_Time(Matrix1; Number; [ResultOrder]; [Selection]; [NumIterations]; [Percentil])`	Marks a series for a batch Holt-Winters forecast, daily (season 7) (run 'Start Batch').
`fMC_Batch_HW_M_Time(Matrix1; Number; [ResultOrder]; [Selection]; [NumIterations]; [Percentil])`	Marks a series for a batch Holt-Winters forecast, monthly (season 12) (run 'Start Batch').
`fMC_Batch_HW_Q_Time(Matrix1; Number; [ResultOrder]; [Selection]; [NumIterations]; [Percentil])`	Marks a series for a batch Holt-Winters forecast, quarterly (season 4) (run 'Start Batch').
`fMC_HWAmodel(alpha; beta; gamma; Period; Matrix1; [Number]; [ResultOrder])`	Holt-Winters forecast (additive) with the given smoothing factors, Number x 1.
`fMC_HWMmodel(alpha; beta; gamma; Period; Matrix1; [Number]; [ResultOrder])`	Holt-Winters forecast (multiplicative) with the given smoothing factors, Number x 1.

Matrix functions

Function	Description
`fMC_PortfolioCombn(AValue)`	Number of portfolio combinations of n titles (without repetition and order).
`fMC_PortfolioMatrix(aMatrix; [AsInput]; [Comparex]; [Comparesx]; [Comparevx])`	All portfolios of the titles: rows (formula summing the MC FLO variables, portfolio name). The Compare arguments are not described in the manual and are ignored.
`fMC_PortfolioCorrMatrix(aMatrix; [FillRandom])`	Correlation formulas for all pairs of titles (random coefficients in +/-0.3 with FillRandom), q x 1.
`fMC_PortfolioCorrMatrixF(aMatrix; [CorrelT])`	All pairwise correlations of simulated variables: rows (name1, name2, coefficient).
`fMC_PortfolioManualMatrix(aMatrix; cMatrix; [isInput])`	Portfolios from a 0-1 matrix (one column per portfolio): rows (formula, name).
`fMC_SimpleMarkovChain(aMatrix; n)`	State distribution after n periods. aMatrix: transition matrix, optionally preceded by a row with the start distribution.
`fMC_NBT(aMatrix; [UseSummary])`	Newcomb-Benford first-digit test (chi-squared), 18 x 3.
`fMC_LDT(aMatrix; [UseSummary])`	Last-digit (before the decimal separator) uniformity test (chi-squared), 19 x 3.

Memory Calculation vector functions

Function	Description
`_simula_Uniform_MC([Lower_Limit]; [Upper_Limit]; [Count])`	Memory Calculation: column of random numbers of the continuous uniform distribution.
`_simula_Triangular_MC([Lower_Limit]; [Upper_Limit]; [Mode]; [Count])`	Memory Calculation: column of random numbers of the triangular distribution.
`_simula_PERT_MC([Lower_Limit]; [Upper_Limit]; [Mode]; [Count])`	Memory Calculation: column of random numbers of the pERT distribution.
`_simula_Normal_MC([Mean]; [StDeviation]; [Count])`	Memory Calculation: column of random numbers of the normal distribution.
`_simula_LogNormal_MC([Shift]; [Mean]; [StDeviation]; [Count])`	Memory Calculation: column of random numbers of the log-normal distribution: Shift + exp(Normal(Mean; StDeviation)).
`_simula_Beta_MC([Param_alfa]; [Param_beta]; [Lower_Limit]; [Upper_Limit]; [Count])`	Memory Calculation: column of random numbers of the beta distribution between Lower_Limit and Upper_Limit.
`_simula_Gumbel_MC([Shift]; [Param_alfa]; [Count])`	Memory Calculation: column of random numbers of the gumbel (extreme value) distribution: location Shift, scale param_alfa.
`_simula_Gamma_MC([Shift]; [Param_alfa]; [Param_beta]; [Count])`	Memory Calculation: column of random numbers of the gamma distribution: Shift + Gamma(scale param_alfa, shape param_beta).
`_simula_Exponential_MC([Shift]; [Param_alfa]; [Count])`	Memory Calculation: column of random numbers of the exponential distribution: Shift + Exponential(mean param_alfa).
`_simula_TruncNormal_MC([Mean]; [StDeviation]; [Lower_Limit]; [Upper_Limit]; [Count])`	Memory Calculation: column of random numbers of the truncated normal distribution.
`_simula_Weibull_MC([Shift]; [Param_alfa]; [Param_beta]; [Count])`	Memory Calculation: column of random numbers of the weibull distribution: Shift + Weibull(scale param_alfa, shape param_beta).
`_simula_chi2_MC([Freedom]; [Count])`	Memory Calculation: column of random numbers of the chi-square distribution.
`_simula_studentt_MC([Freedom]; [Count])`	Memory Calculation: column of random numbers of the student-t distribution.
`_simula_Pareto_MC([Shift]; [Param_alfa]; [Count])`	Memory Calculation: column of random numbers of the pareto distribution with minimum Shift (> 0) and shape param_alfa.
`_simula_Fisher_MC([Freedom1]; [Freedom2]; [Count])`	Memory Calculation: column of random numbers of the f distribution.
`_simula_ArcSin_MC([Lower_Limit]; [Upper_Limit]; [Count])`	Memory Calculation: column of random numbers of the arcsine distribution.
`_simula_Cauchy_MC([Shift]; [Param_alfa]; [Count])`	Memory Calculation: column of random numbers of the cauchy distribution: location Shift, scale param_alfa (the cell shows the median).
`_simula_Frechet_MC([Start]; [Shift]; [Param_alfa]; [Count])`	Memory Calculation: column of random numbers of the frechet distribution: minimum Start, scale Shift, shape param_alfa.
`_simula_Logistic_MC([Location]; [Shift]; [Count])`	Memory Calculation: column of random numbers of the logistic distribution: location Location, scale Shift.
`_simula_DiscreteUniform_MC([Lower_Limit]; [Upper_Limit]; [Count])`	Memory Calculation: column of random numbers of the discrete uniform distribution (integers).
`_simula_Bernoulli_MC([Probability]; [Count])`	Memory Calculation: column of random numbers of the bernoulli distribution: 1 (TRUE) with the given probability, else 0.
`_simula_Binomial_MC([Probability]; [Tries]; [Count])`	Memory Calculation: column of random numbers of the binomial distribution: successes in Tries trials.
`_simula_Poisson_MC([Occurency]; [Count])`	Memory Calculation: column of random numbers of the poisson distribution.

Function	Description
`_simula_Pascal_MC([Probability]; [Success]; [Count])`	Memory Calculation: column of random numbers of the pascal (negative binomial) distribution: failed attempts until Success successes; Probability = probability of a failure.
`_simula_Geometric_MC([Probability]; [Count])`	Memory Calculation: column of random numbers of the geometric distribution: failures before the first success.
`_simula_Hypergeometric_MC([Tries]; [Population]; [Success_Possibles]; [Count])`	Memory Calculation: column of random numbers of the hypergeometric distribution (draws without replacement).
`_simula_calculation_MC([Vector1]; [Vector2]; MMOperation)`	Memory Calculation: row-wise operation on two vectors (+ - * / */+ */- //+ //- Max Min com csme inc const).
`_simula_correlation_MC([Vector1]; [Vector2]; CorrCoef; [CorrType]; [DoSort])`	Memory Calculation: correlates two vectors (rank reordering), n x 2.
`_simula_Bounded_MC([Vector1]; LowerBound; UpperBound; [Number])`	Memory Calculation: keeps the values of Vector1 within [LowerBound, UpperBound].

Appendix B Files and names

File or name	Purpose
`Documents\MC_FLO_language.txt`	the chosen language (shared by both editions)
`Documents\MC_FLO_API.txt`	the Anthropic API key of the AI mentor
`Documents\MC_FLO_mentor.txt`	the mentor's answer language and your instructions
`FLO_i_ / FLO_o_ / FLO_c_ <name>`	named ranges of inputs, outputs and correlations
`FLO_r_<name>`	stored results of an output (Use DIP)
`MonteCalc_…`	hidden names in an Excel workbook holding the settings (LibreOffice: document properties)
`<model>_dd_mm_yyyy_hh_mm_ss_MC_FLO.ods / .xlsx`	saved result workbook

On Linux, `Documents` is the Documents folder in your home directory.